

Corrected Audit Packet for Strongly Rosenthal Compacta

Run `fa_banach_001`: corrected partial

June 18, 2026

Abstract

This packet replaces the previous separable-case packet. That proof is not valid. There are two independent failures: pointwise-convergent Baire-one sequences may still be equivalent to the ℓ_1 unit-vector basis in the supremum norm, and the proposed Banach-space quotient does not collapse the fibers of the admissible cover. The valid replacement is a topological embedding of the target compactum into a dual space E^* , where

$$E = \overline{\text{span}}^{\|\cdot\|} \{ \alpha(e(x)) : x \in X \} \subset V^*.$$

This is useful structure, but it is not enough to prove admissibility because it does not identify E as the dual of a separable Rosenthal Banach space. The original separable strongly-Rosenthal-versus-admissible problem therefore remains open in this packet.

1 Source question and corrected status

The source is Glasner–Megrelishvili, *Representations of dynamical systems on Banach spaces not containing ℓ_1* , arXiv:0803.2320 [1]. It defines strongly Rosenthal compacta and admissible compacta and asks whether the two classes coincide.

continuity property when restricted to B^* . Equivalently, every restriction of v^{**} to a bounded subset M is fragmented as a function $(M, w^*) \rightarrow \mathbb{R}$ (see Fact 4.11 below).

Answering a question of Talagrand [42, Problem14-2-41], R. Pol [36] gave an example of a separable compact Rosenthal space K which cannot be embedded in $\mathcal{B}_1(X)$ for any compact metrizable X . We say that a compact space K is *strongly Rosenthal* if it is homeomorphic to a subspace of $\mathcal{B}_1(X)$ for a compact metrizable X . We say that a compact space K is *admissible* if there exists a metrizable compact space X and a bounded subset $Z \subset C(X)$ such that the pointwise closure $\text{cls}_p(Z)$ of Z in $\mathbb{R}^X$ consists of Baire 1 functions and $K \subset \text{cls}_p(Z)$. Clearly every admissible compactum is strongly Rosenthal. We do not know whether these two classes of compact spaces coincide.

Corrected run status. The previous active packet claimed to prove that every separable strongly Rosenthal compactum is admissible. That claim should not be used. The present packet records why the proof fails and keeps the strongest valid replacement obtained from the audit.

Outcome. Corrected partial only. No proof, counterexample, or published resolution of the original question is recorded here. The active mathematical content is the embedding theorem of Section 5 and the two failure mechanisms in Sections 3 and 4.

2 Definitions and source tools

Let X be compact metrizable. A bounded family $F \subset C(X)$ is called Rosenthal when its pointwise closure satisfies

$$\text{cls}_p(F) \subset \mathcal{B}_1(X).$$

Equivalently, by the Rosenthal–Talagrand criterion used in the source, no sequence in F is equivalent to the unit-vector basis of ℓ_1 .

A compactum K is strongly Rosenthal if it embeds into $\mathcal{B}_1(X)$ for some compact metrizable X . It is admissible if it embeds into $\text{cls}_p(F)$ for a bounded Rosenthal family $F \subset C(X)$ on a compact metrizable space.

The previous packet used two source results.

- Theorem 6.3 of [1]: for a Rosenthal family $F \subset C(Y)$, the compactum $L = \text{cls}_p(F)$ admits a representation in V^{**} for a separable Rosenthal Banach space V , with evaluation functionals $\alpha(y) \in V^*$.
- Theorem 6.16 of [1]: admissibility can be characterized via weak-star compact subsets of second duals of separable Rosenthal spaces.

3 First failure: pointwise convergence does not prevent ℓ_1

The previous proof assumed that a pointwise-convergent sequence in $\mathcal{B}_1(X)$ cannot be equivalent, in the supremum norm, to the unit-vector basis of ℓ_1 . This is false.

Proposition 3.1 (A compact metrizable counterexample). *There is a compact metrizable space X and a pointwise-null sequence $(f_n) \subset \mathcal{B}_1(X)$ such that (f_n) is isometric to the unit-vector basis of ℓ_1 in the supremum norm.*

Proof. Let

$$X = \{\infty\} \cup \bigcup_{m=1}^{\infty} \{m\} \times \{-1, 1\}^m$$

be the one-point compactification of the countable discrete union of the finite sign cubes. This is compact metrizable. Define

$$f_n(\infty) = 0, \quad f_n(m, \sigma) = \begin{cases} \sigma_n, & n \leq m, \\ 0, & n > m. \end{cases}$$

For each fixed point of X , f_n is eventually zero, hence $f_n \rightarrow 0$ pointwise. Since X is countable compact metrizable, every real-valued function on X is Baire-one; indeed every inverse image of an open interval is countable and therefore F_σ .

For scalars $a_1, \dots, a_N$, choose $m = N$ and $\sigma_n = \text{sgn}(a_n)$. Then

$$\left| \sum_{n=1}^N a_n f_n(m, \sigma) \right| = \sum_{n=1}^N |a_n|.$$

The reverse inequality is the triangle inequality. Thus

$$\left\| \sum_{n=1}^N a_n f_n \right\|_{\infty} = \sum_{n=1}^N |a_n|.$$

□

Corollary 3.2. *The graph-coordinate construction in the previous packet need not produce a Rosenthal family. Pointwise compactness of the target family in $\mathcal{B}_1(X)$ is not enough to rule out ℓ_1 -sequences in the corresponding supremum-norm coordinate family.*

4 Second failure: the quotient does not collapse fibers

The previous packet then attempted to repair the admissible cover by quotienting the representing Banach space. In the notation of Theorem 6.3, let

$$\nu_0 : L \rightarrow V^{**}, \quad \alpha : Y \rightarrow V^*$$

be the representation, and let $e : X \rightarrow Y$ denote the graph embedding. The old proof set

$$N = \bigcap_{x \in X} \ker \alpha(e(x)) \subset V$$

and claimed that the quotient map $q : V \rightarrow V/N$ makes $q^{**}\nu_0(h)$ depend only on the fiber of the restriction map $\pi : L \rightarrow K$.

There are two problems.

1. In the actual construction behind Theorem 6.3, the vectors represented by the graph coordinates generate the relevant predual space. For the graph set, the above N is therefore typically just $\{0\}$. Then the quotient does not collapse any fibers.
2. The annihilator step uses the wrong closure. If $A = \{\alpha(e(x)) : x \in X\} \subset V^*$, then

$$\left(\bigcap_{\phi \in A} \ker \phi \right)^\perp = \overline{\text{span}}^{w^*} A$$

in V^* , by the bipolar theorem for the dual pair (V, V^*) . But an element of V^{**} is norm-continuous on V^* , not generally weak-star continuous. Therefore vanishing on A gives vanishing on the norm-closed span of A , not on its weak-star closed span.

Thus the quotient proof cannot prove that K is weak-star compact in the second dual of a separable Rosenthal quotient.

5 The valid replacement

The norm-closure issue points to a genuine theorem. It gives a topological embedding into a dual Banach space, but not the second-dual representation needed for admissibility.

Theorem 5.1 (Norm-closed evaluation-span embedding). *Let L be an admissible cover of a compactum K , with continuous surjection $\pi : L \rightarrow K$, arising from a graph restriction construction. Assume L is represented as in Theorem 6.3 of [1] by*

$$\nu_0 : L \rightarrow V^{**}, \quad \alpha : Y \rightarrow V^*,$$

where V is separable Rosenthal and

$$h(y) = \langle \nu_0(h), \alpha(y) \rangle \quad (h \in L, y \in Y).$$

Put

$$E = \overline{\text{span}}^{\|\cdot\|} \{ \alpha(e(x)) : x \in X \} \subset V^*.$$

Then the formula

$$\sigma(\pi(h)) = \nu_0(h)|_E$$

defines a topological embedding

$$\sigma : K \hookrightarrow E^*$$

when E^* carries its weak-star topology from the predual E .

Proof. If $\pi(h_1) = \pi(h_2)$, then

$$h_1(e(x)) = h_2(e(x)) \quad (x \in X).$$

Equivalently,

$$\langle \nu_0(h_1) - \nu_0(h_2), \alpha(e(x)) \rangle = 0 \quad (x \in X).$$

Since $\nu_0(h_1) - \nu_0(h_2) \in V^{**}$ is norm-continuous on V^* , it vanishes on the norm-closed linear span E . Hence $\nu_0(h_1)|_E = \nu_0(h_2)|_E$, so σ is well-defined.

Conversely, if the restrictions to E agree, then they agree on every $\alpha(e(x))$. Therefore $h_1(e(x)) = h_2(e(x))$ for all x , which is precisely $\pi(h_1) = \pi(h_2)$ in the graph-restriction quotient. Thus σ is injective.

Continuity follows because, for each $u \in E$, the function

$$h \mapsto \langle \nu_0(h), u \rangle$$

is continuous on L after first checking it on the generating $\alpha(e(x))$'s and then passing to the norm closure uniformly on bounded sets. Therefore the factor map $K \rightarrow E^*$ is weak-star continuous. A continuous injection from compact K into the Hausdorff space E^* is a homeomorphism onto its image. $\square$

Remark 5.2. *Theorem 5.1 is the strongest valid replacement found in this audit. It does not prove admissibility because it gives $K \subset E^*$, not $K \subset W^{**}$ for a separable Rosenthal Banach space W . To apply Theorem 6.16 of the source, one would need to identify E with W^* , or otherwise represent E^* as such a second dual in a way compatible with the embedding. No such identification follows from the construction.*

6 Literature and remaining problem

The audit supplied for this correction reports no published resolution of the underlying open statement. It also reports that later work continued to treat the separable strongly Rosenthal case as unresolved: a later Glasner–Megrelishvili survey still describes the separable strongly Rosenthal case as unknown, Marciszewski–Pol formulate an equivalent envelope problem and solve only the Aleksandrov–Urysohn special case, and specialist material from 2021 continues to distinguish the relevant classes.

Problem 6.1. *Does every separable strongly Rosenthal compactum admit an admissible representation?*

Problem 6.2. *Can the space $E = \overline{\text{span}}^{\|\cdot\|} \{ \alpha(e(x)) : x \in X \} \subset V^*$ from Theorem 5.1 be represented, under natural hypotheses, as the dual of a separable Rosenthal Banach space?*

Problem 6.3. *Can one build a strongly Rosenthal compactum for which the embedding of Theorem 5.1 cannot be lifted to a second-dual representation over any separable Rosenthal space?*

7 What this packet supersedes

This packet supersedes the previous active packet in this folder. The old packet is preserved under

`history/invalid_previous_packet_2026.06.18/`

only as provenance. Its claimed theorem that every separable strongly Rosenthal compactum is admissible should not be cited as a result of this run.

8 Human review checklist

1. Check Proposition 3.1: the compact space is metrizable, the functions are Baire-one, and the ℓ_1 estimate is exact.
2. Check that the old graph-coordinate argument really needs a no- ℓ_1 conclusion in supremum norm and therefore fails.
3. Check the annihilator closure distinction: weak-star closure is supplied by the bipolar theorem, while the fiber argument requires norm closure because elements of V^{**} are norm-continuous on V^* .
4. Check that Theorem 5.1 is well-defined and is a genuine topological embedding into E^* .
5. Check that this embedding is insufficient for admissibility without an additional theorem identifying E as a Rosenthal dual predual.

References

- [1] E. Glasner and M. Megrelishvili, Representations of dynamical systems on Banach spaces not containing ℓ_1 , arXiv:0803.2320.
- [2] H. P. Rosenthal, A characterization of Banach spaces containing ℓ_1 , *Proc. Nat. Acad. Sci. USA* **71** (1974), 2411–2413.